Bedfordview Primary School
TERM 4 Year End Test

Name: Memo	Grade 5 : A B C D
Subject: Natural Science and Technology	
Examiner: Mrs C. Nash	Date: 22 November 2024
Moderator: Mr J. Nash	Time Allocation: 1hour 30min Total: 60
INSTRUCTIONS 1. Read all the instructions and questions carefully. 2. Write neatly and legibly. 3. Write in full sentences unless otherwise instructed. 4. Look carefully at the mark allocation when giving your answers. 5. Answer in the spaces provided. This paper consists of 6 pages.	Parent's signature & comments: _____ _____ _____

1	2	3	4	5	6	7
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

	60% Section A: Term 3 Energy & Electricity and Energy & Movement					40% Section B: Term 4 Planet Earth and Surface of the Earth				
Question	**1**	**2**	**3**	**4**	**5**	**1**	**2**	**3**	**Total**	**%**
Allocated mark	7	6	12	4	7	7	5	12	60	
Learner's mark										
Moderated mark										

Content: **Energy and Electricity**: cells and batteries; mains electricity. **Energy and Movement**: wheel and axle; elastics and springs. **Planet Earth**: the earth moves and **Surface of the Earth**
Skills: Understanding and interpreting data on storing of energy, circuits, electricity and transferring of energy. Identifying objects that are elastic. Understanding stored energy and movement of energy

SECTION A – Energy and Electricity & Energy and Movement /36

Question 1: Definitions (Knowledge recall) /7

Match the words in column A with their meaning in column B.

Column A	Column B
A. Components	1. Things you do to stop something dangerous happening.
B. Battery	2. Smallest whole part of a battery.
C. Precautions	3. Groups of related parts that work together as a whole for a particular purpose.
D. Systems	4. The parts that make up an electric circuit.
E. Cell	5. Kinetic energy.
F. Circuit	6. Group of cells.
G. Movement	7. A system that transfers electrical energy.

A-4	B-6	C-1	D-3	E-2	F-7	G-5

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Questions 2: (True or False) **/6**

State whether the following statements are true or false.

2.1) Energy is stored in cells/batteries.

True

2.2) An electrical circuit is a source of stored energy.

False

2.3) A simple circuit consists of a cell and connecting wires.

False

2.4) Mains electricity is electricity from the power station that is transferred by the national grid to homes and schools.

True

2.5) A power station can use coal as a source of energy to produce electricity.

True

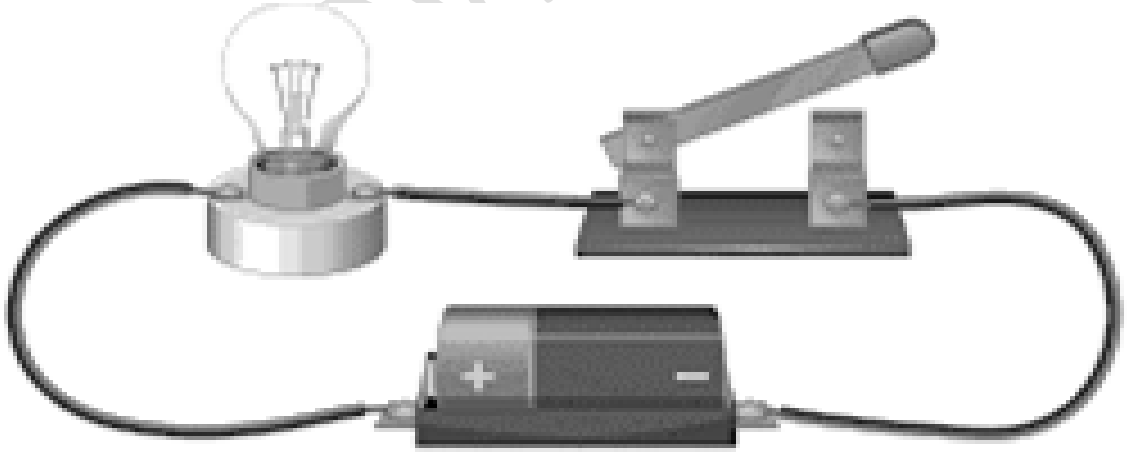
2.6) Safey precautions should not be taken into consideration when you work with electricity.

False

Question 3: Electrical Circuit. **Total = 12**

Study the electrical circuit below and answer the questions that follow.

3.1 **Explain the function** of each component in the following electrical circuit: **/4**



Battery: Source of electrical energy

Switch: To connect and disconnect the power supply from battery to light bulb

Light bulb: To shine when the circuit is connected properly.

Copper wire: Transfer the flow of electrical energy to where it is needed.

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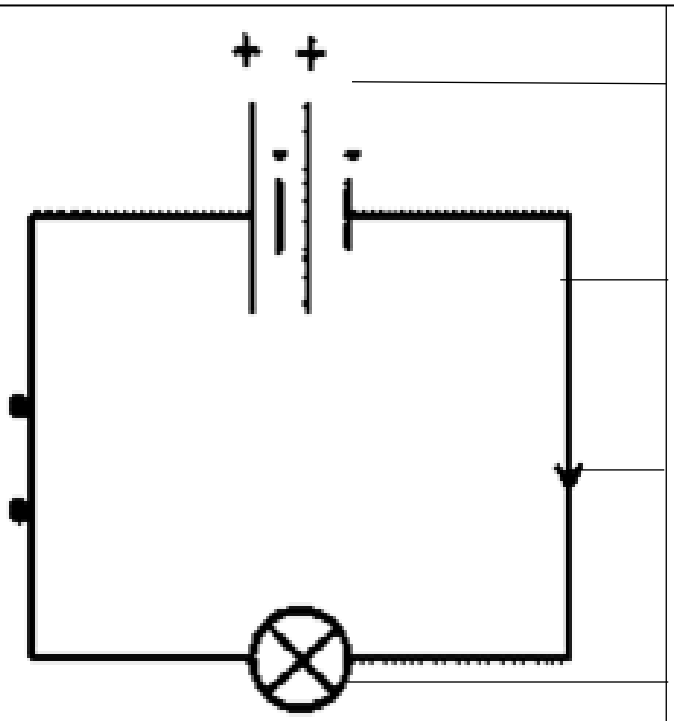
3.2 **Draw and label** a diagram of a closed simple circuit so that the globe will shine. **8 x ½ = /4**

- cell - light bulb - connecting wire - direction of energy flow



- Cell
- Copper/connecting wire
- Direction of energy flow
- Light bulb

3.3 **Match** the word to the correct description. **/4**

Words	Description
1) Output Energy	a) a substance that can be eaten or burned to produce energy.
2) Fuels	b) saying what you think might happen.
3) Input Energy	c) energy such as heat and light that we get from burning fuel.
4) Predict	d) the amount of energy that must be added to fuel to make it burn.

1)____c____ 2)____a____ 3)____d____ 4)____b____

Question 4: True or False **/4**

State whether the following sentences are true or false.

a) The wheel is one of the oldest inventions. True

b) Using a vehicle allows us to move around more easily. True

c) The bulb in an electrical circuit will shines when it is connected to the cells by one wire.
False

d) In a torch circuit, electrical energy is transferred from the light bulb to the cell.
False

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Question 5: Mains electricity /7

Place the following statements in their correct order:

How electricity flows from power stations to our homes, schools, and offices.	Order 1-7	
1. Substations reduce high-voltage electricity to a lower voltage, which can be used in homes and offices.	6	3
2. An electrical appliance draws the energy needed to work from the mains supply.	3	7
3. Powerlines carry electricity from the power stations to substations in areas where people live and work.	1	2
4. Wall sockets built into the walls of a building provide places for people to safely plug into the mains supply.	7	5
5. When plugged into the wall socket, an appliance is connected to the mains electrical supply.	4	6
6. Power stations burn fuels such as coal to generate electricity.	5	1
7. Electricity boxes at home control the supply of electricity to the building.	2	4

SECTION B – Planet Earth and Beyond / 24

Question 1: Definitions (Knowledge recall) /7

Match the words in column A with their meaning in column B.

Column A	Column B
A. Orbit	1. The hard layer on the outside of the Earth.
B. Rotation	2. Smal pieces of broken-down rock.
C. Conserve	3. The pathway that one object takes around another.
D. Crust	4. When dead plants and animals break down and become part of the soil.
E. Structure	5. To use wisely and to avoid wasteful or destructive use.
F. Decompose	6. The movement of an object around itself.
G. Sediments	7. How different parts are put together.

A-3	B-6	C-5	D-1	E-7	F-4	G-2

Questions 2: (True or False) /5

State whether the following sentences are true or false.
If false, rewrite the statement to make it true.

a) The Earth's axis is vertical. False

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The Earth's axis is slightly tilted.

b) The whole Earth can receive light at the same time. False

Only the part of the Earth facing the sun can receive light at a given time.

c) The side of the Earth that receives light is experiencing night. False

The side of the Earth that receives light is experiencing daylight.

d) The Sun takes 365 ¼ days to revolve around the Earth. False

The Earth takes 365 ¼ days to revolve around the Sun.

e) The shape of the Earth's orbit is square. False

The shape of the Earth's orbit is oval or sphere.

Question 3: Different layers of Earth's crust. **Total = /12**

3.1) **Complete** the following paragraph by filling in the missing words. **6 x ½ = /3**

Rocks are big pieces of solid material. Rocks break down to make smaller rocks and sand . Rocks form the bottom layer of land. The topsoil and the subsoil both sit on top of a layer of rocks .

3.2) **Draw and label** a diagram showing the different layers of Earth's crust. **/6**

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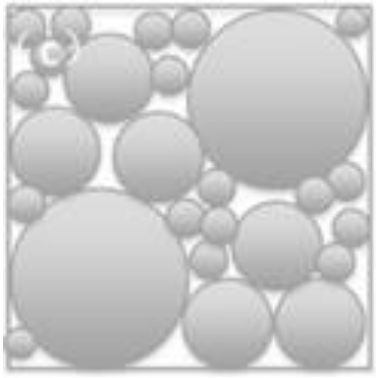
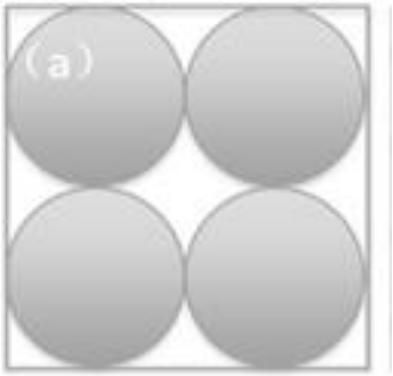
Diagram	Labels
	TOPSOIL SUBSOIL ROCK

3.3 **Complete** the following table by filling in the missing information. **3 x ½ = /1½**

	Clay	Loam	Sand
Size of particles	Most of the particles in clayey soil are small	Loam contains sand and clay particles.	Most of the particles in sandy soil are quite large
Drainage	Clayey soil holds a lot of water	Loam soil lets some water pass through	Sandy soil does not hold water well.
Nutrients	Clayey soil has some	Loam contains the right amount of nutrients	Sandy soil contains very few nutrients

3.4 **Draw** an illustration of each soil type below.
Use the correct colour and pay attention to particle size. **3 x ½ = /1½**

Clay	Loam	Sand
		

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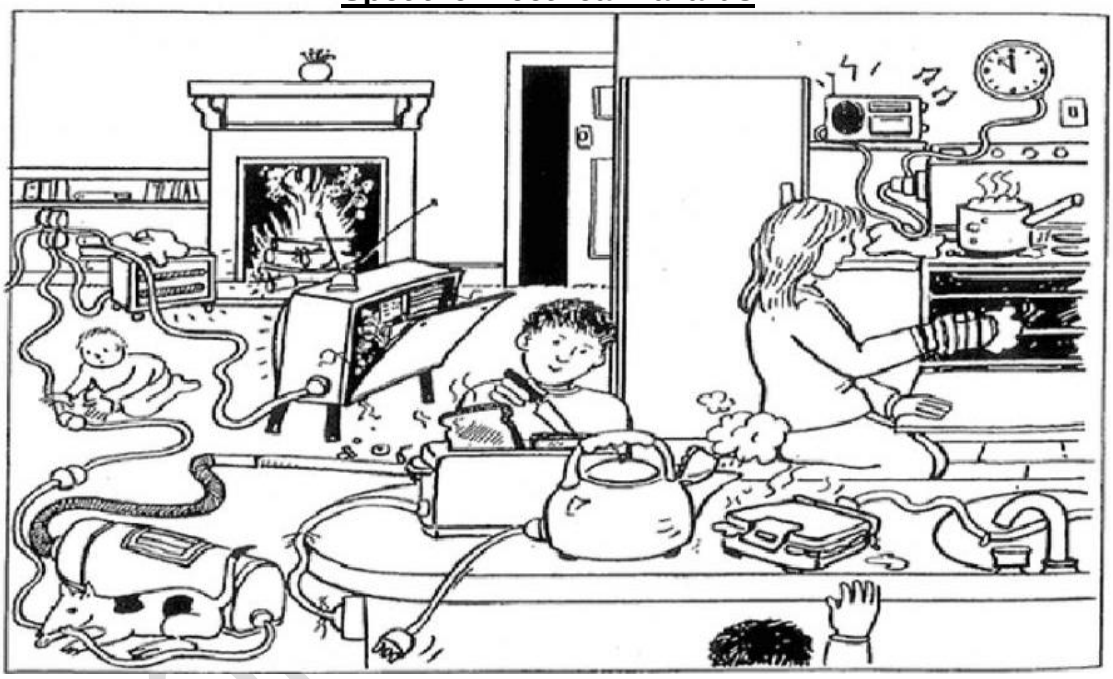
Extension Activity

To be completed after you have checked **all your answers** and only if you have finished.

Spot the Electrical Hazards



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